OTOMOTO TORONTO MOTO INC.
99 River Street, Toronto, ON M5A 3P4, Canada

Bike link

Two boxes. A sketch of the hardware, what the finished boxes look like, and what to buy first.

BIKE BOX · BUILD THIS FIRST

Under the seat

One box on one motorcycle. It wakes when the key turns on, asks the bike for kilometers and fault codes, then goes back to sleep. After the bike has been sitting, it measures the resting battery voltage and keeps that reading until a phone or the shop tablet is next to it.

The first handoff is Bluetooth. Staff read the reading on the phone and type it into the motorcycle page. Power is a fused wire on the battery. The red diagnostic plug does not power the box.

MAILBOX · LATER

In a window

A listener with wall power and the building’s Wi-Fi. It uses the same short-range radio as the bike box, once that radio is fitted. At the shop it sits in a plastic box in a window at 99 River Street that faces where the bikes park. Not a metal electrical room. Not a closed cabinet.

A phone is the first handoff. The mailbox is the second. Bike-to-bike relay, and coverage across Toronto, are out of scope.

IN THIS PACKET

1. What the bike box and the mailbox look like.
2. Wiring, and the red 6-pin, pins 1 through 6.
3. Parts for the first Bluetooth box.
4. Mailbox parts, only after the sleep test.
5. The pass mark, and the three numbers the box stores.

No cell. No satellite.

No cellular modem, no SIM card, and no monthly radio bill. No satellite radio and no satellite fee. Customer texts stay on the Wix Inbox thread the shop already uses. Nothing in this build sends a text.

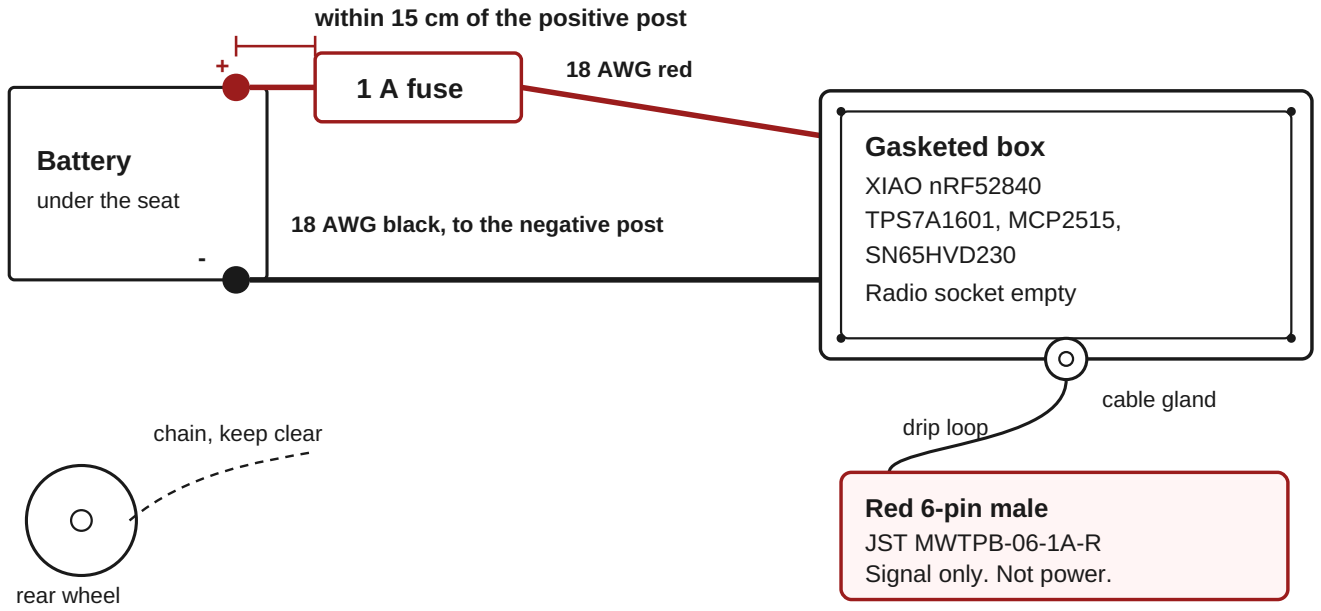
Build the Bluetooth box first. Leave the radio off until that box sleeps, reads a bike, and a person can copy the reading into the motorcycle page.

What it looks like

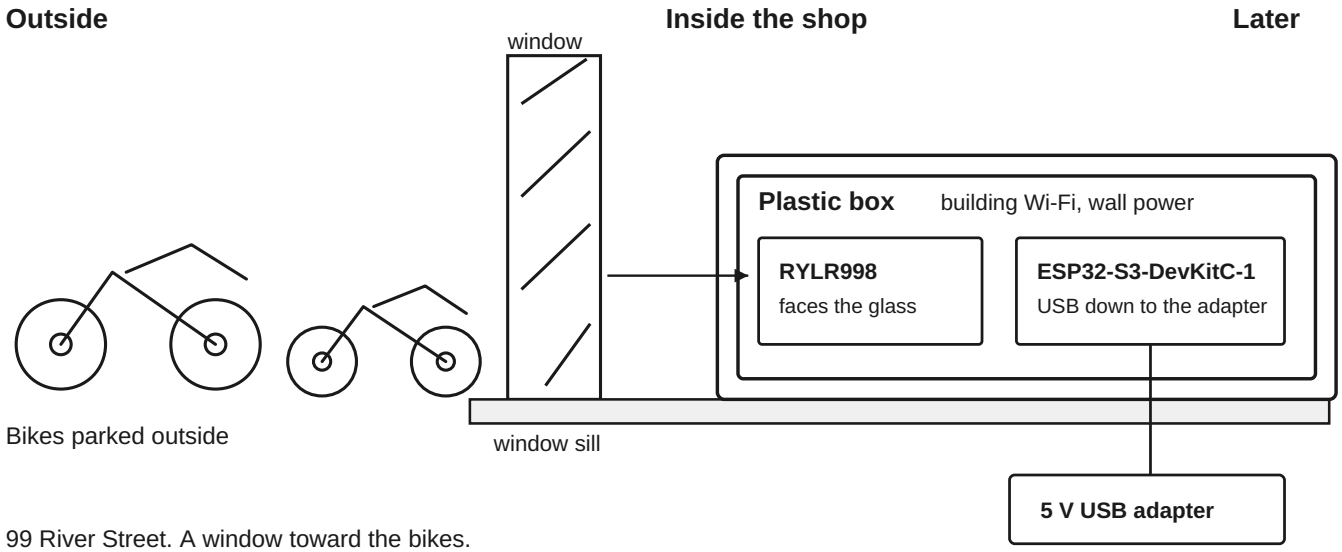
Build sketches, not a product render. The bike box is the one to build now. The mailbox waits.

Seat The box straps in the space under this seat.

Build this one first. Bluetooth only. Radio socket stays empty.



Bike box under the seat. Not to scale. The 1 A fuse is at the battery, on the red 18 AWG wire. The small gasketed box holds the XIAO. The red 6-pin is the diagnostic plug. It does not feed power.



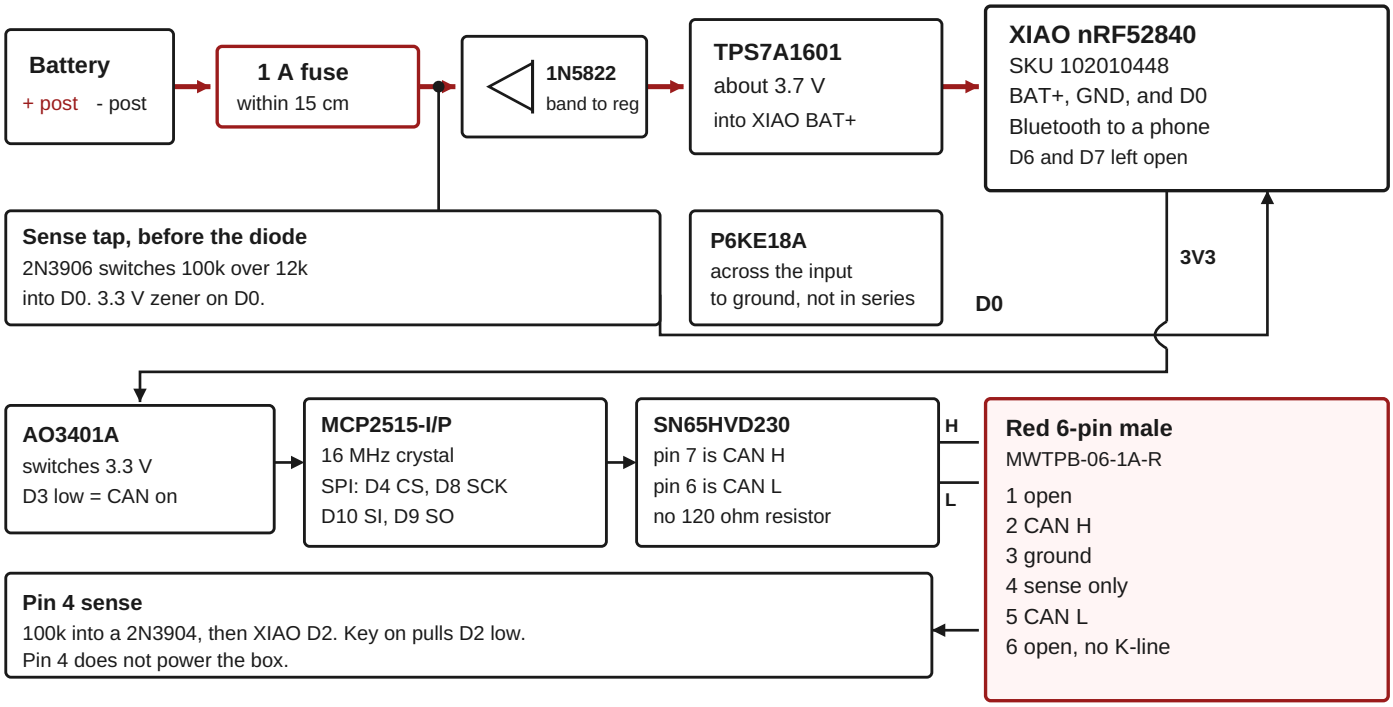
99 River Street. A window toward the bikes.

Not a metal room. Not a closed cabinet.

Mailbox, later. Not to scale. Plastic box on a window sill. The RYLR998 antenna is inside the module, so the whole mailbox has to sit in the window. A roof antenna is a different radio, and this sheet does not name one.

Wiring

First build, Bluetooth only. The radio header stays empty. The fuse is within 15 cm of the battery positive post. The red plug is signal only.



Ground Battery negative, regulator pad, XIAO GND, transceiver pin 2, and plug pin 3.

1N5822 band points at the regulator. P6KE18A band points at the fused 12 V.

Not a painted frame bolt. First build: radio header empty.

Power path and the red plug. Not to scale. Thick red is battery power. Thin lines are signal. D6 and D7 stay open until a radio is fitted.

Male housing JST MWTPB-06-1A-R. The bike already has the red female housing. The pigtail is 22 AWG. The 18 AWG battery leads do not go through this plug. K-line is not this build. Do not force this plug into a Harley.

Pin	What the plug is for	What this cable does
1	Left to the motorcycle maker	Leave open
2	CAN H, when the bike uses CAN	Twisted pair to SN65HVD230 pin 7
3	Ground	Signal ground, 22 AWG, to the box
4	Battery positive. It may be always on, or it may follow the key.	Sense only. 100 kΩ into a transistor. Does not power the box.
5	CAN L, when the bike uses CAN	Twisted pair to SN65HVD230 pin 6
6	K-line on some other bikes	Leave open

Before the crimps, battery negative disconnected: pin 3 reads through to battery negative, pin 4 is near 0 V with the key off and about battery voltage with the key on, and pins 2 and 5 are not shorted to ground or to battery positive. If pin 4 stays hot with the key off, leave it open and move the sense wire to a circuit that really follows the key.

What to buy

First prototype, Bluetooth only. Leave the radio socket empty. If a distributor is out, buy the spec in the build notes. Do not substitute a different kind of part.

Part	Why	Qty
Seeed Studio XIAO nRF52840, SKU 102010448	Bluetooth and the log, on one small board. Buy the plain board. The Sense and the Plus do not match this wiring.	1
Texas Instruments TPS7A1601	Makes about 3.7 V for the XIAO battery pads from motorcycle voltage.	1
210 kΩ 1%, and 100 kΩ 1%	Set the regulator to about 3.70 V. One more 100 kΩ 1% is the top of the voltage divider.	1 × 210 kΩ 2 × 100 kΩ 1%
10 µF ceramic, X7R, 25 V or more, and 0.1 µF ceramic	Two 10 µF caps on the regulator. One 0.1 µF across each CAN chip.	2 × 10 µF 2 × 0.1 µF
1N5822	Schottky after the fuse, so the box cannot push voltage back up the battery wire.	1
P6KE18A	Spike clipper across the regulator input. A normal charging voltage leaves it alone. If charging sits above 15 V, use a P6KE20A instead.	1
1 A ATM (mini) blade fuse and an inline holder	Protects the new wire at the battery. The box itself draws well under 100 mA.	1
18 AWG red and 18 AWG black	Battery lead and main ground. Black goes to the battery negative post.	1 each
Microchip MCP2515-I/P	CAN controller. The XIAO has no CAN. DIP-18, runs on 3.3 V. An SOIC MCP2515 is the same chip if you can solder it.	1
16 MHz crystal and two load capacitors	Clock for the MCP2515. A crystal marked 18 pF usually wants about 22 pF on each leg.	1 crystal 2 caps
SN65HVD230DR	3.3 V CAN transceiver. Order the DR reel. The older SN65HVD230D is obsolete. Do not buy a TJA1050 or an MCP2551.	1
SOIC-8 to DIP adapter	Solder the transceiver once and plug it in on the bench.	1
AO3401A	Turns 3.3 V on to the CAN chips only while the box is reading. Off while parked.	1
2N3904 and 2N3906	Ignition sense, and the switch for the voltage divider.	2 × 2N3904 1 × 2N3906
100 kΩ, 12 kΩ 1%, 47 kΩ, 10 kΩ	Divider bias, ignition sense, CAN gate pull-up, and the CAN reset pull-up. Keep the 12 kΩ at 1%.	5 × 100 kΩ 1 × 12 kΩ 1% 1 × 47 kΩ 2 × 10 kΩ
3.3 V zener, 1N5226B or BZX55C3V3	Clamps the voltage-sense pin if a spike hits while the divider is on.	1
JST MWTPB-06-1A-R, terminals SMWTWS-A021T-064, seals A03WS-MWT-N	Male red 6-pin. Pins 2, 3, 4, and 5 only. Pins 1 and 6 stay empty. The terminals are easy to ruin, so a few extras are worth having.	1 plug 4 terminals 4 seals
22 AWG twisted pair, plus a ground and a sense wire	CAN H, CAN L, signal ground, and ignition sense. These wires do not power the box.	1 set
Gasketed ABS box and a cable gland	Under-seat weather and wash water. A Hammond 1591 is the kind of box. The exact shell is not important.	1
Perfboard, solder, heat-shrink, zip ties	What you assemble it on. Strap the board down. Do not pot the box.	shop stock

A fixed 3.3 V regulator is the wrong voltage for the XIAO battery pads. On the bike, solder the MCP2515 or strap the socket so vibration cannot walk the chip out.

Mailbox parts, later

Buy these only after the Bluetooth box sleeps under 50 microamps at the fuse, and a person has typed a reading into the motorcycle page.

Part	Why	Qty
Reyax RYLR998	Short-range radio, 915 MHz, antenna built in. One goes in the bike box. One goes in the mailbox. Power it from 3.3 V. Cross TX and RX.	2
Espressif ESP32-S3-DevKitC-1	Mailbox brain. Wi-Fi, USB, and a 3.3 V pin that can run the radio. It sits on wall power.	1
5 V USB power adapter, 1 A	Mailbox power. A normal phone charger is enough.	1

The radio must be set to 0 dBm before the first transmission, and this PDF is not a certification.

Put the mailbox in the plastic box in the window from the sketch. The module’s own setting when it arrives is much hotter than 0 dBm. Change it before you transmit. The shop command in the build notes is AT+CRFOP=0.

Pass mark and what it stores

Under 50 microamps

Measure in series at the fuse. Key off. USB unplugged. CAN power off. Wait one minute after the box goes to sleep. Over 50 microamps, it stays on the bench. Do not put it on a customer bike.

On a small 8 amp-hour sport-bike battery, 50 microamps all winter is about 3 percent. A box at 1 milliamp is not acceptable.

Three stored values

1. **Kilometers.** The whole number already on that motorcycle, in kilometres or miles. The box does not convert. If the bike does not answer, store a blank. Never send 0 to mean “could not read it.”
2. **Resting battery voltage.** One decimal, about 20 minutes after the key is off. A reading while the engine is running is not stored. If there is no rested sample, leave the voltage blank. The box does not call the battery good or flat. A lithium battery rests higher than a lead-acid battery. Send the measured number anyway.
3. **Fault codes.** Short codes such as P0335 or C14, up to 20. Store them. Do not clear them.

A reading needs at least one of those three. Staff type it into the motorcycle page. The sticker and the Bluetooth name are TM-BL- plus four digits.

What the box does not do

- It does not decide that a service is due. The shop app does that from the kilometers and the date.
- It does not clear codes, and it does not send a clear-codes command.
- It does not text the customer. Customer texts stay on Wix Inbox.